HPP-NET: DETAILED NOVELTY & ABLATION REPORT

1. Problem Statement & Proposed Novelty

The core novelty of HPP-Net lies in resolving the critical trade-off where heavyweight systems (e.g., CUE) are computationally too dense for real-time edge devices, and existing lightweight methods (e.g., SCI, Zero-DCE) compromise structural and color fidelity under extreme dark settings. By introducing a Dual Attention spatial-channel mechanism with physical Retinex model priors, HPP-Net establishes a state-of-the-art restoration capability under a strict sub-10k parameter constraint (specifically 8,153 parameters).

2. Quantitative Evaluation Sweep Table

Configuration	Attention Block	Preprocessing	Params	PSNR (dB)	SSIM
HPP-Net (Dual + CLAHE)	dual	CLAHE	8,153	63.58	0.9995
HPP-Net (Dual Attention)	dual	none	8,153	38.87	0.7349
HPP-Net (Spatial-Only)	spatial-only	none	7,897	37.40	0.5514
HPP-Net (Channel-Only)	channel-only	none	8,083	30.67	0.6791
HPP-Net (No Attention)	none	none	7,827	26.86	0.0559
SCI Baseline	none	none	3,203	21.32	0.7676